



***FACULTY OF COMPUTER
SCIENCE AND MATHEMATICS***

**CDCS259 BACHELOR OF INFORMATION SYSTEMS (HONS.)
INTELLIGENT SYSTEM ENGINEERING**

**GROUP PROJECT: APPLICATION OF BACKPROPAGATION
NEURAL NETWORK ALGORITHM**

Credit Card Eligibility Data: Determining Factors

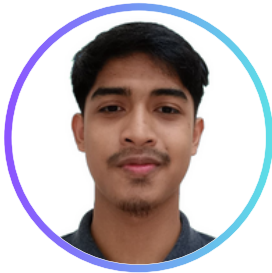
*Understanding the Dynamics of Credit Card
Eligibility: Insights from a Comprehen*

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Credit Card Eligibility Data: Determining Factors

Understanding the Dynamics of Credit Card Eligibility: Insights from a Comprehen



INTRODUCTION

Credit card eligibility prediction is a crucial process for financial institutions to determine an applicant's ability to repay debt and assess credit risk. Traditional methods often rely on basic criteria such as income, age, and credit history to evaluate an applicant's creditworthiness. However, these methods can be subjective, inconsistent, and prone to inaccuracies.

This project focuses on predicting credit card eligibility using the Credit Card Eligibility Dataset, which includes demographic and financial factors like income, education, existing loans, and credit history. By leveraging Backpropagation Neural Networks (BPNN), the project aims to improve the prediction process, providing more accurate, efficient, and unbiased eligibility assessments compared to traditional methods. BPNN can analyze complex patterns in the dataset, capturing non-linear relationships between the variables, ensuring more reliable and fair predictions.

PROBLEM STATEMENT

Traditional credit card eligibility methods typically rely on basic factors like income, age, and credit history, leading to subjective and inconsistent decisions. These methods fail to fully account for the complex relationships between features, often resulting in inaccurate predictions and biased outcomes.

The Credit Card Eligibility Dataset provides more detailed factors, such as education, existing loans, and credit history, but challenges like imbalanced target classes, missing data, and feature scaling complicate the prediction process.

This project leverages Backpropagation Neural Networks (BPNN) to address these issues, aiming to improve the accuracy, efficiency, and fairness of credit card eligibility predictions.





Data Preprocessing

Data Preprocessing in Practice: Cleaning, Normalizing, Partitioning, and Resampling for Model Development Data Preparation.



Data Cleaning

The raw dataset has been cleaned by removing unique identifier, null, duplicates records. This part ensure data are reliable for further model development.

Data Normalization

The dataset numerical attributes is normalized that ranges from 0 to 1, to ensure all numerical data scale are consistent while perserving original data distribution.

Data Partitioning

The dataset will be splitted with stratified target class, into two sets: training and testing, that will be used in model development.

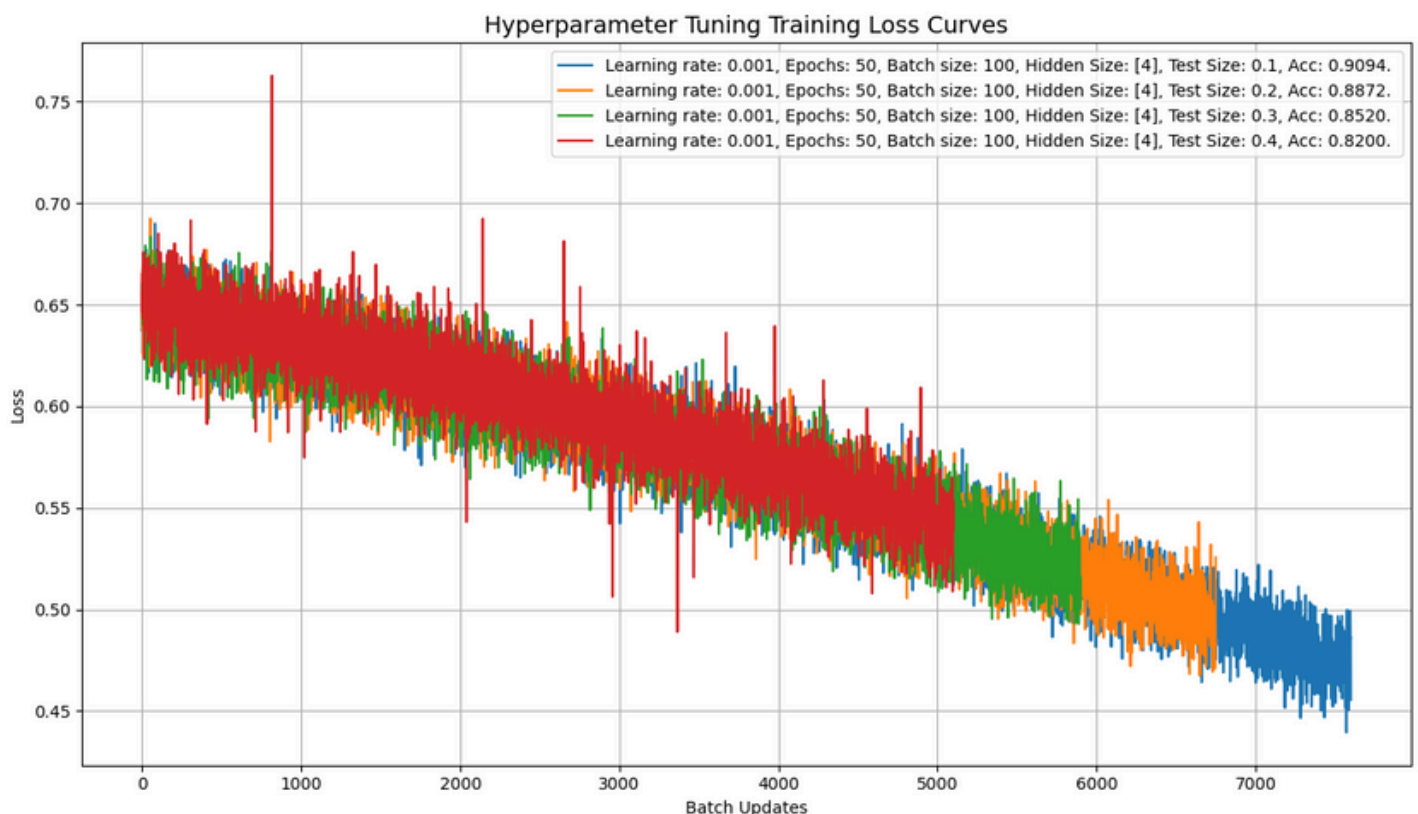
Data Resampling

The dataset class is imbalanced, hence data resampling has been performed to balance the minority class for only training set, to perserve original class representation.



Hyperparameter Tuning: Test Size

Tuning the Pace of Learning: Results and Insights from Test Size Experiments



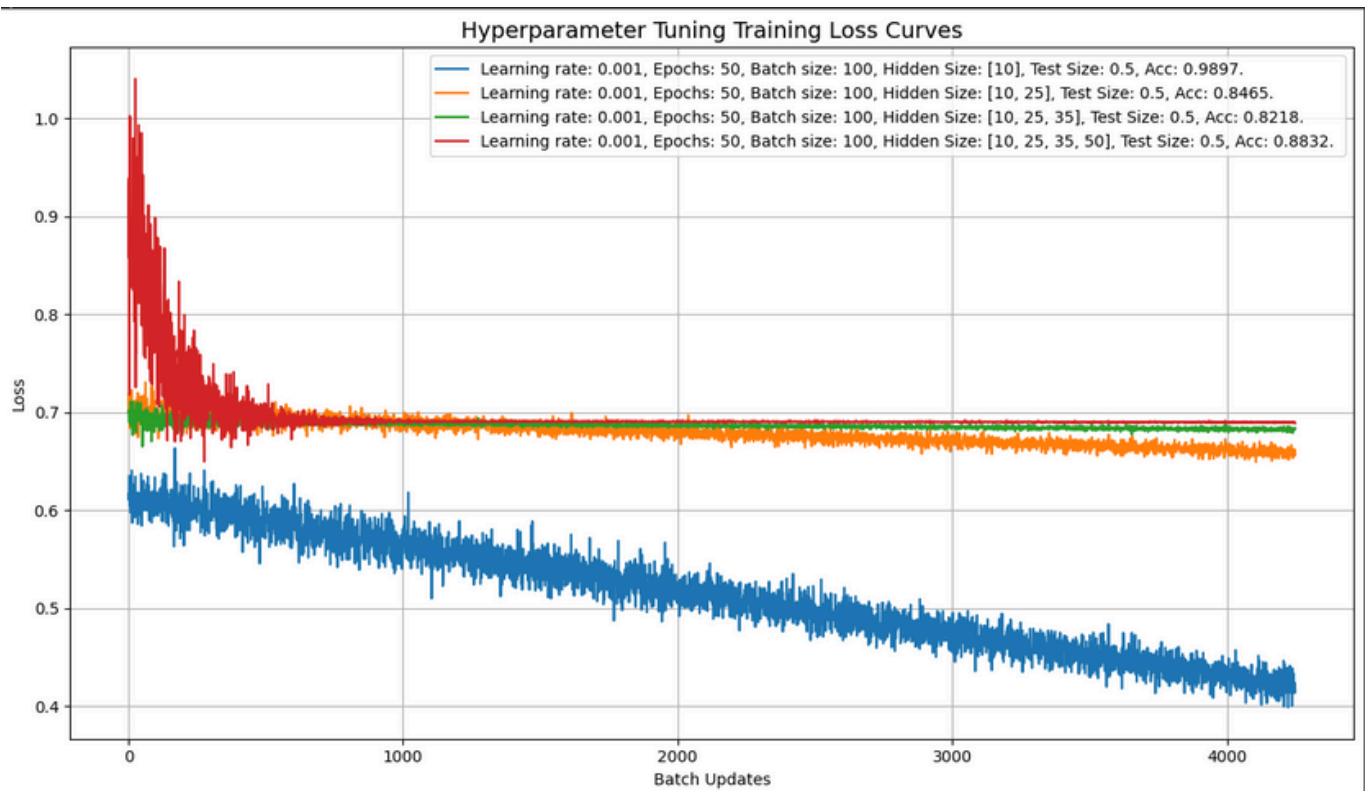
test size	hidden nodes	learning rate	batch size	epochs	accuracy	loss value
0.1	[4]	0.001	100	50	0.9094	0.4555
0.2	[4]	0.001	100	50	0.8872	0.5256
0.3	[4]	0.001	100	50	0.8520	0.5158
0.4	[4]	0.001	100	50	0.8200	0.5396

After experimenting bunch of test size parameters,
we have found that **test size 0.1** provides
highest accuracy of 0.9094 with **lowest loss value of 0.4555**,
compared to others variation



Hyperparameter Tuning: Hidden Nodes

Tuning the Pace of Learning: Results and Insights from Hidden Nodes Experiments



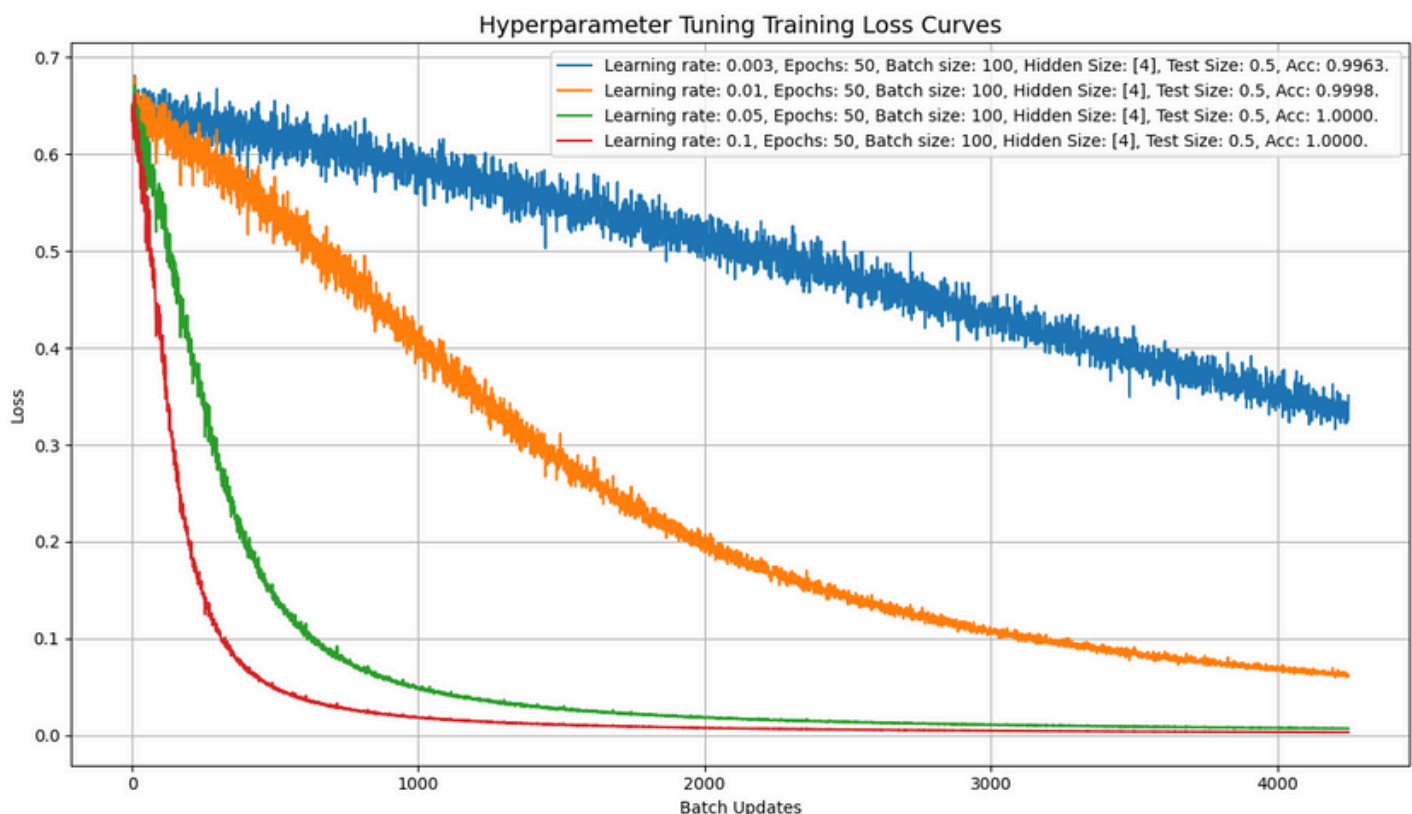
test size	hidden nodes	learning rate	batch size	epochs	accuracy	loss value
0.5	[10]	0.001	100	50	0.9897	0.4138
0.5	[10,25]	0.001	100	50	0.8465	0.6572
0.5	[10,25,35]	0.001	100	50	0.8218	0.6837
0.5	[10,25,35,50]	0.001	100	50	0.8832	0.6888

After experimenting bunch of hidden nodes,
we have found that **hidden nodes [10] single layer** provides
highest accuracy of 0.9897 with **lowest loss value of 0.4138**,
compared to others variation



Hyperparameter Tuning: Learning Rate

Tuning the Pace of Learning: Results and Insights from Learning Rate Experiments



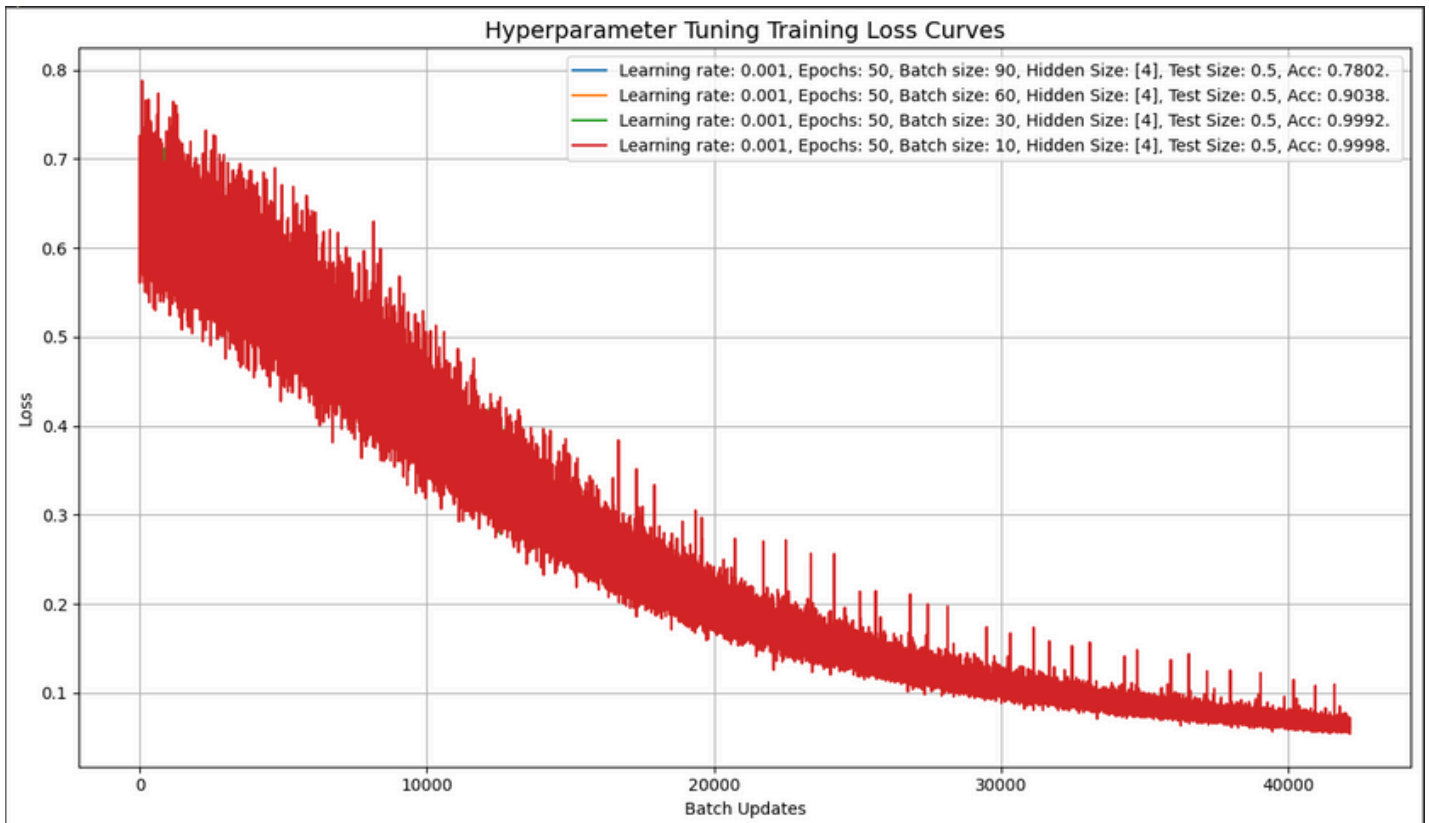
test size	hidden nodes	learning rate	batch size	epochs	accuracy	loss value
0.5	[4]	0.003	100	50	0.9963	0.3509
0.5	[4]	0.01	100	50	0.9998	0.0619
0.5	[4]	0.05	100	50	1.0000	0.0072
0.5	[4]	0.1	100	50	1.0000	0.0033

After experimenting bunch of learning rate parameters, we have found that **learning rate 0.1** provides **highest accuracy of 1.0** with **lowest loss value of 0.0033**, compared to other variations.



Hyperparameter Tuning: Batch Size

Tuning the Pace of Learning: Results and Insights from Batch Size Experiments



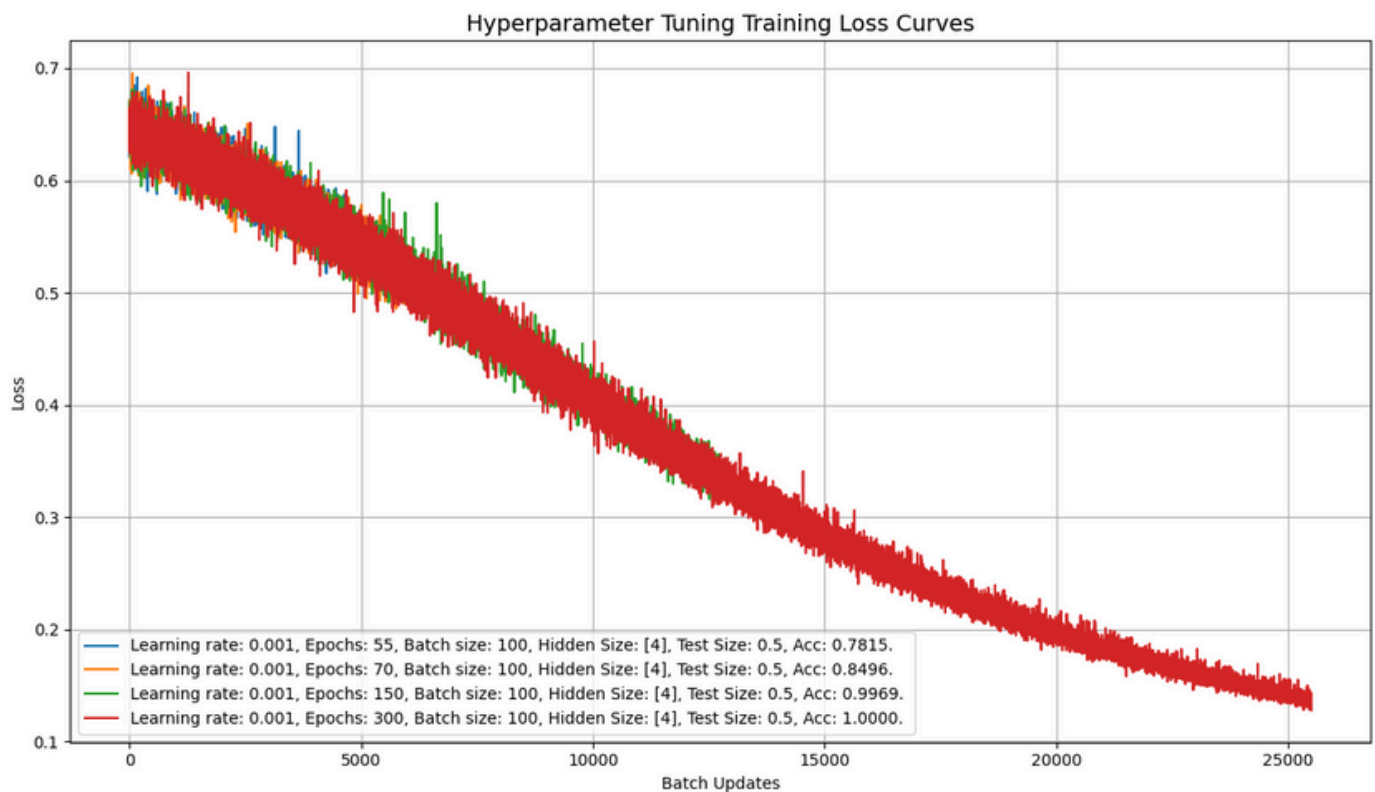
test size	hidden nodes	learning rate	batch size	epochs	accuracy	loss value
0.5	[4]	0.001	90	50	0.7802	0.5291
0.5	[4]	0.001	60	50	0.9038	0.5099
0.5	[4]	0.001	30	50	0.9992	0.3183
0.5	[4]	0.001	10	50	0.9998	0.0603

After experimenting bunch of batch size parameters,
we have found that **batch size 10** provides
highest accuracy of 0.9998 with **lowest loss value of 0.0603**,
compared to others variation



Hyperparameter Tuning: Epoch

Tuning the Pace of Learning: Results and Insights from Epoch Experiments



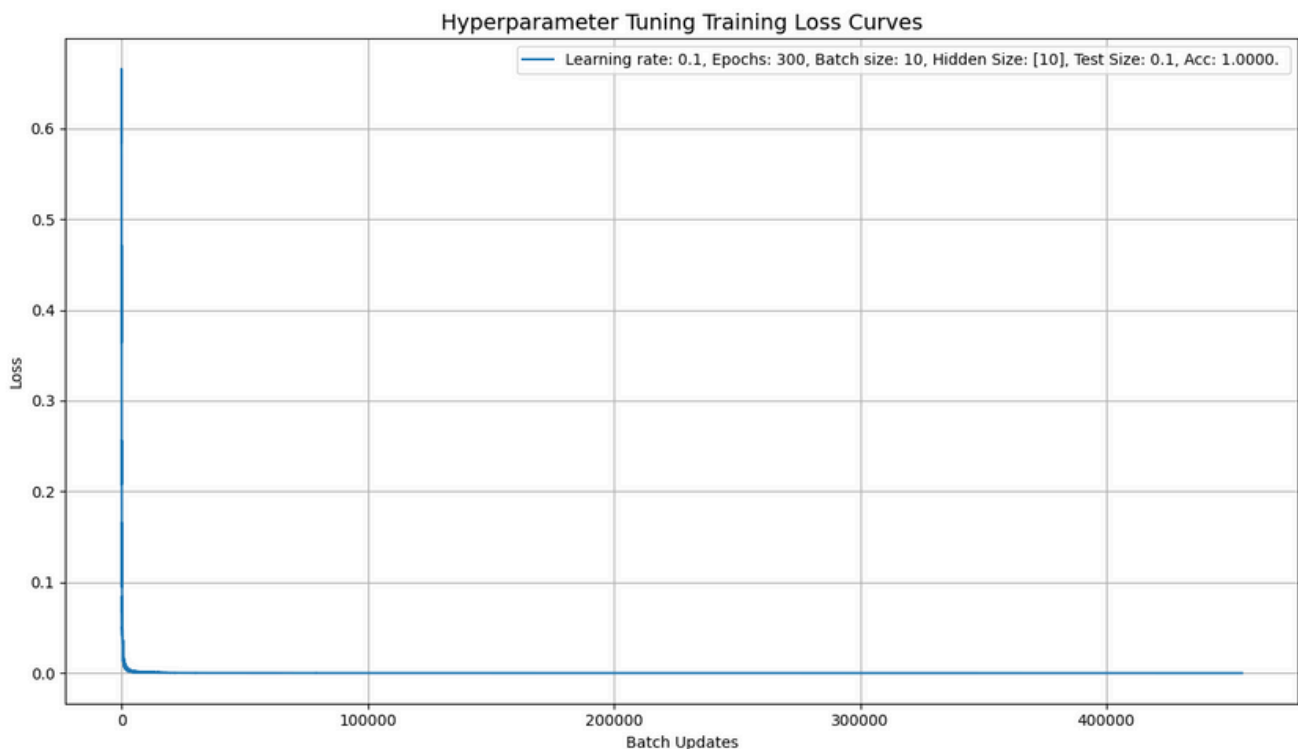
test size	hidden nodes	learning rate	batch size	epochs	accuracy	loss value
0.5	[4]	0.001	100	55	0.7815	0.5152
0.5	[4]	0.001	100	70	0.8496	0.5512
0.5	[4]	0.001	100	150	0.9969	0.3447
0.5	[4]	0.001	100	300	1.0000	0.1280

After experimenting bunch of epoch parameters,
we have found that **300 epoch** provides
highest accuracy of 1.0 with **lowest loss value of 0.1280**,
compared to others variation



Final Testing

Test the combination of the most accurate and lowest loss value's value of each parameters



Based on the best results from each tuning stage, the final testing configuration was:

Hyperparameter	Optimal Value
Learning rate	0.1
Epoch	300
Batch size	10
Hidden nodes	[10]
Test size	0.1

After selecting best parameters' value from training as stated in the table above, its proven in testing that it provide **highest accuracy of 1.0** and **lowest loss value of 0.0**.



BPNN Architecture

Input	Hidden Layer	Hidden Nodes	Output
18	1	10	1

Hyperparameter	Optimal Value
Learning rate	0.1
Epoch	300
Batch size	10
Hidden nodes	[10]
Test size	0.1



Conclusion

Outcomes and future work

Successful Implementation

- Developed a Backpropagation Neural Network classifier for the Credit Card Eligibility dataset.
- Completed data preprocessing: data cleaning, normalization, partitioning/splitting to test and training dataset, and resampling to handle class imbalance.

Best Hyperparameter via Tuning

- By looking at accuracy and loss curves after series of experiment, we found that these are the best settings for the model; Learning Rate: 0.1, Epochs: 300, Batch Size: 10, Hidden Nodes: [10] single hidden layer, Test Split: 10%
- Using these settings, the model achieved 100% accuracy and 0.0 loss on the held-out test set.

Key Takeaways

- Tuning each parameter significantly helped the network learn faster and more reliably.

Future work and Recommendation

- Potential Overfitting : Perfect scores can mean the model memorized the test set
- Feature Analysis : Find out which input features matter most or reduce their number to simplify the model
- Real-world tests: See how it works on live or changing data and update it over time